Acknowledgements

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Dark Patterns:
Introduction

Contents

- What are Dark Patterns?
- Why are they called that?
- Why should I care?

What are Dark Patterns?

- Some websites use tricks to make people agree to share more data than they intended to, or to spend more money than they planned to.
- User Interface experts got together and categorized a number of these tricks that are commonly found on websites, and called them *Dark Patterns*.

What are Dark Patterns?

- Gray, *et al.* (2018) defined them as "*instances where designers use their knowledge of human behavior (e.g., psychology) and the desires of end users to implement deceptive functionality that is not in the user's best interest*"

- Gray, C.M., Kou, Y., Battles, B., Hoggatt, J. and Toombs, A.L., 2018, April. The dark (patterns) side of UX design. In Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (pp. 1-14).

Why are they called that?

- Dark Patterns get their name from an existing concept in computer science called *Design Patterns*.
- Design patterns are a series of general, reusable solutions to a commonly occurring computer programming problems.
- They generally refer to the style of the computer program itself, whereas Dark Patterns focus on the Interface.

Why should I care?

- Dark Patterns deceptively get people to give their data and money away.
- You or a family member are vulnerable to these misleading interfaces. No matter how careful you are, some are extremely effective.

Dark Patterns:
Types of Patterns

Types of Dark Patterns

- Trick questions
- Sneak into Basket
- Roach Motel
- Privacy Zuckering
- Price Comparison Prevention
- Misdirection
- Hidden Costs
- Bait and Switch
- Confirmshaming
- Disguised Ads
- Forced Continuity
- Friend Spam

https://darkpatterns.org/types-of-dark-pattern.html

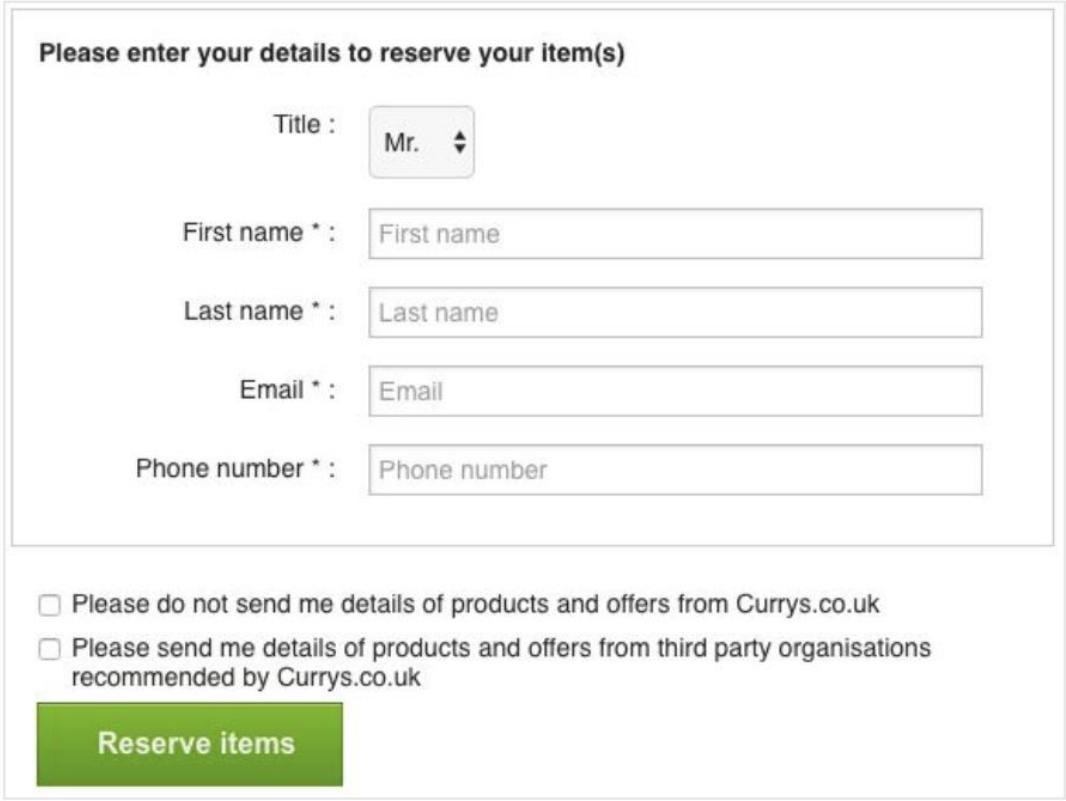
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DARK PATTERN

Trick Questions

Trick Questions

- This is very common when registering with a service. Typically a series of checkboxes is shown, and the meaning checkboxes is alternated so that ticking the first one means "opt out" and the second means "opt in".

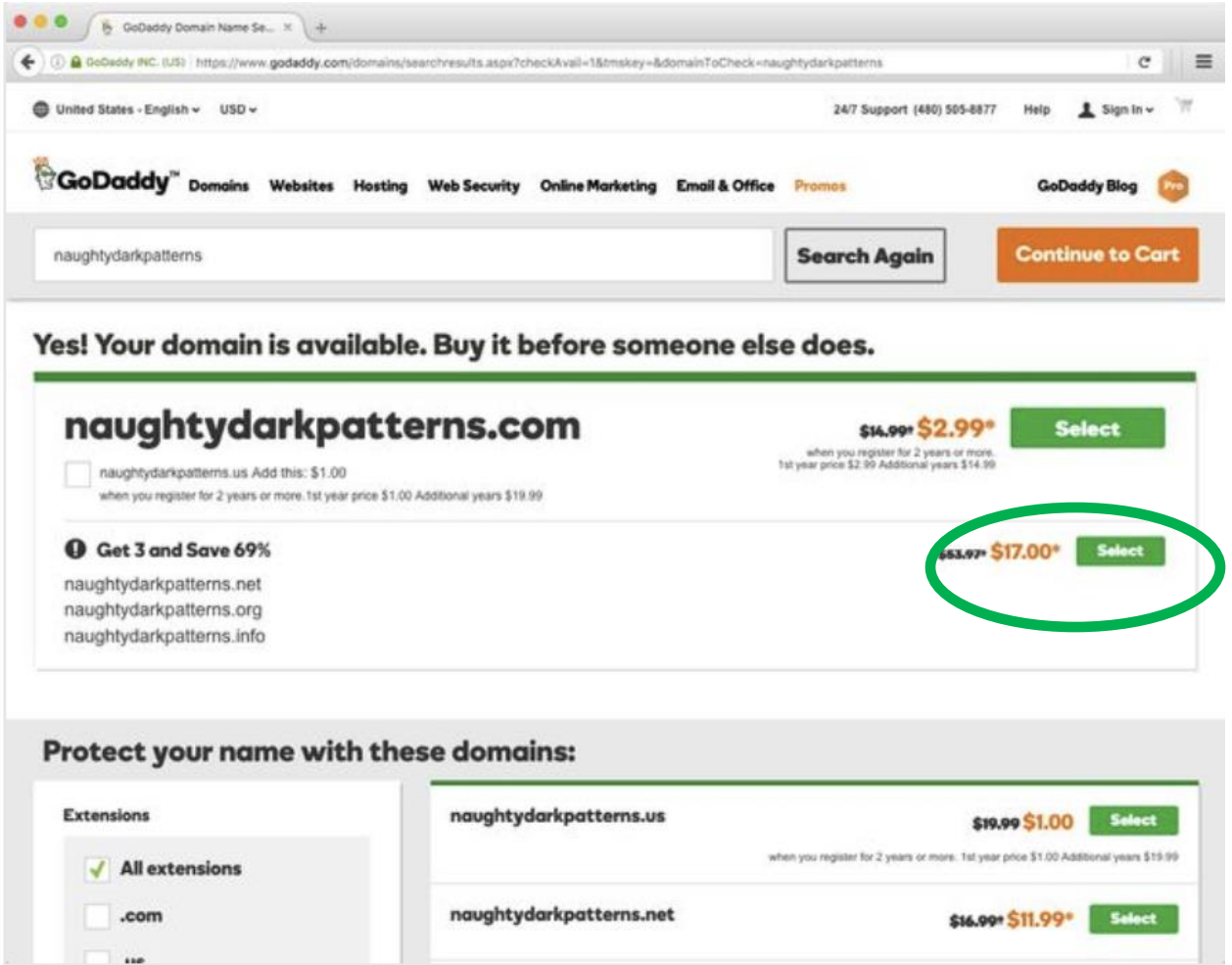




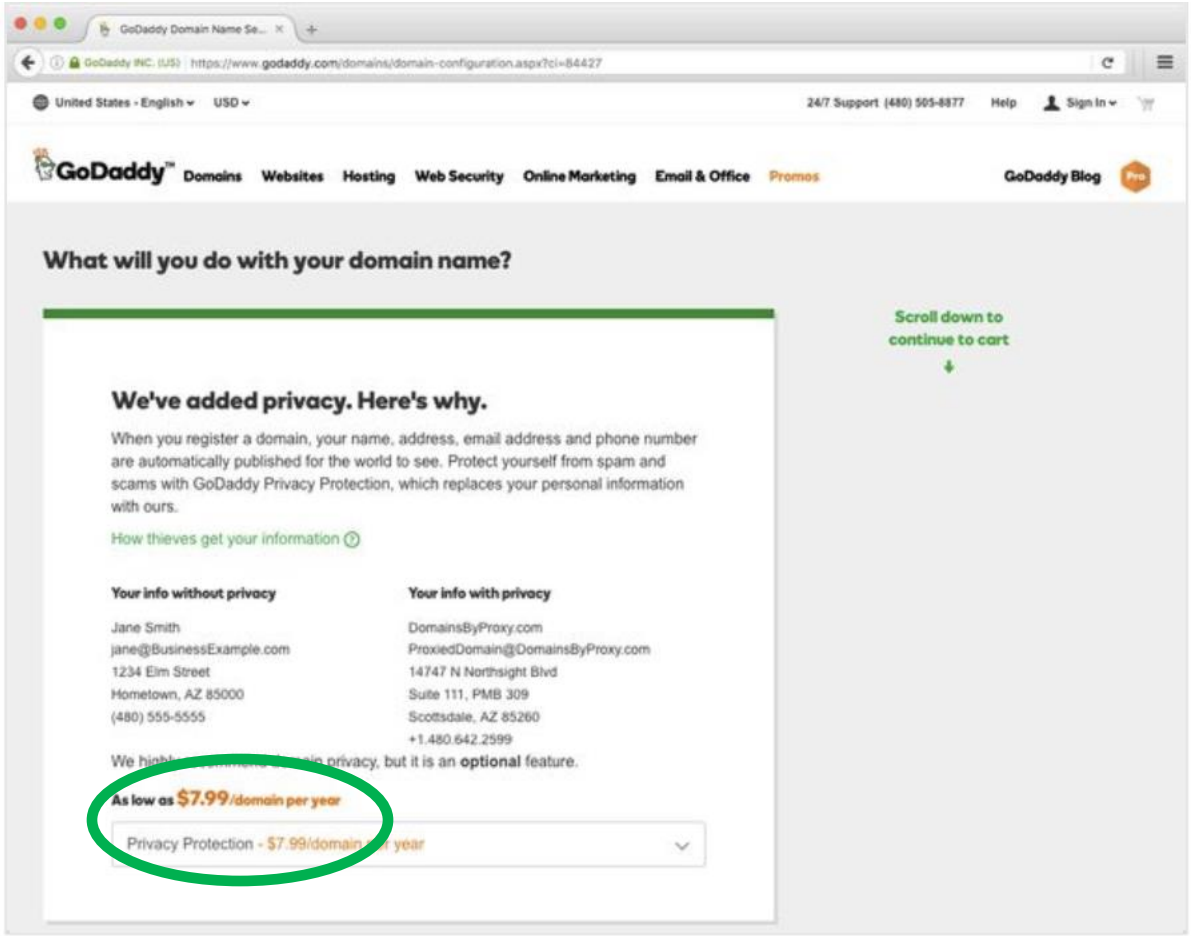
Sneak into Basket

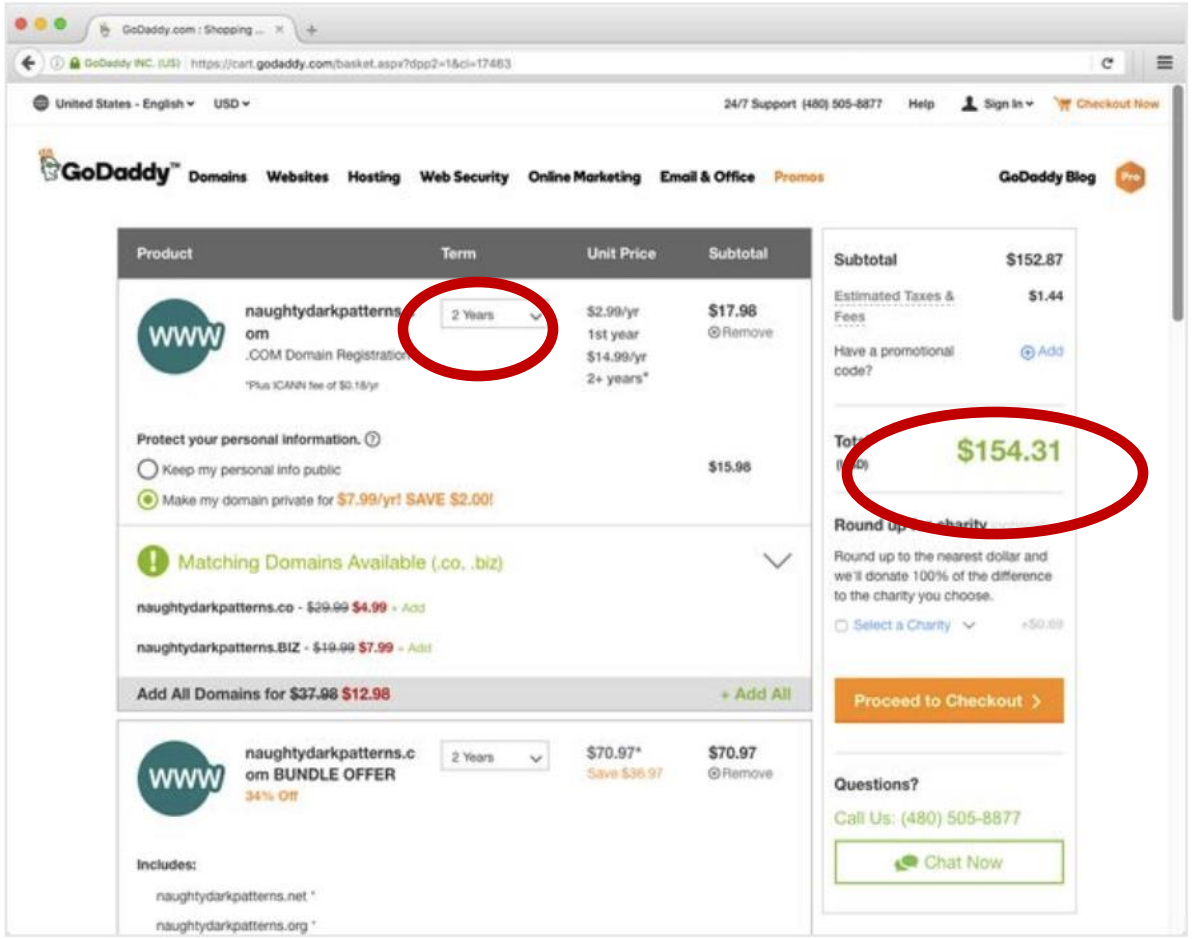
Sneak into Basket

- You attempt to purchase something, but somewhere in the purchasing journey the site sneaks an additional item into your basket, often through the use of an opt-out radio button or checkbox on a prior page.



Sneak into Basket



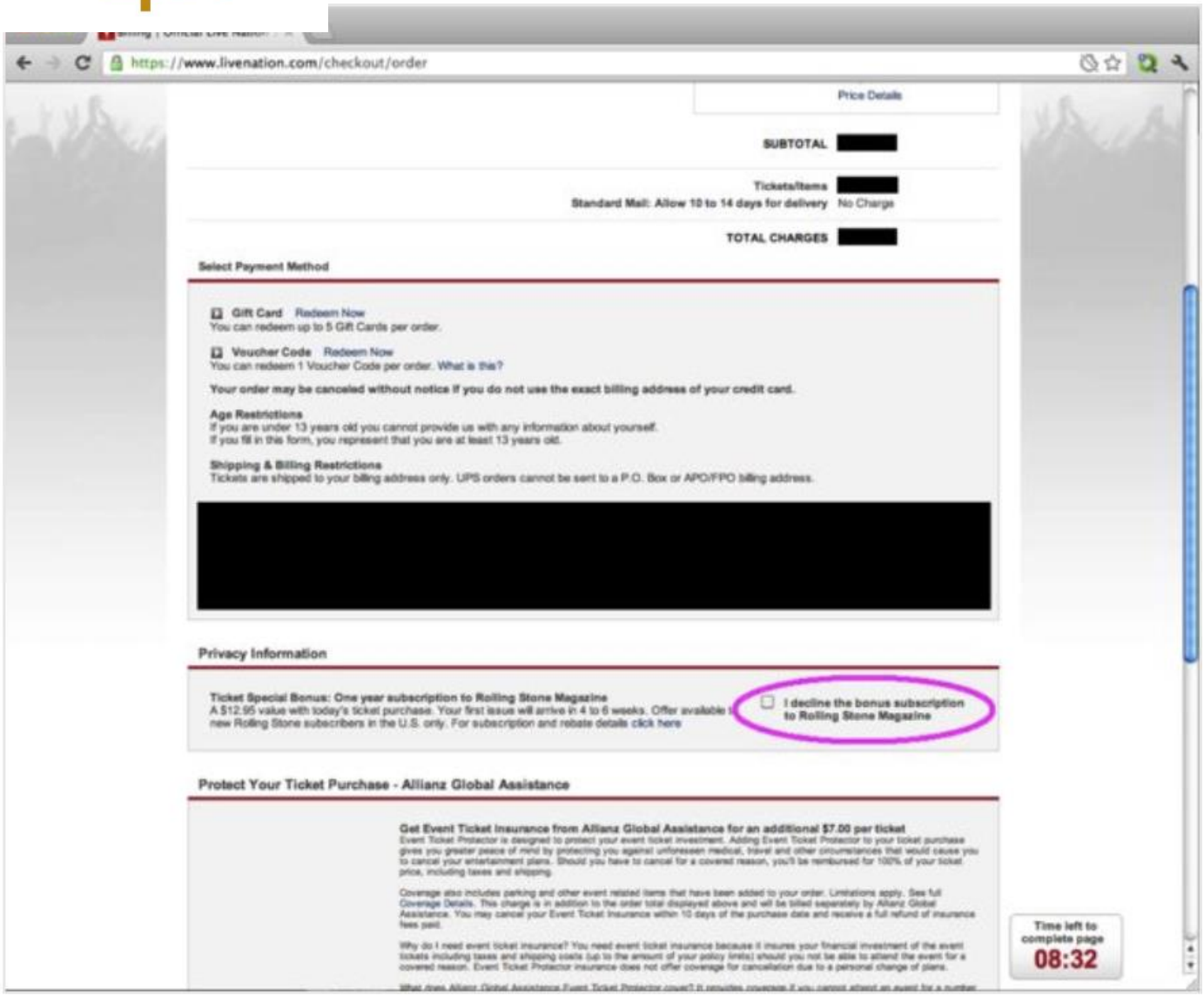


2 years was "sneaked into basket". Costing of multiple domain names was unclear from the outset.



Roach Motel

Roach Motel



- When purchasing tickets, *livenation.com* tries to sneak a subscription to a magazine (**into your basket** via a **trick question** on the checkout page, shown in screenshot.
- Note that the design is "opt out" - the purchase is made for you automatically unless you notice the checkbox and realise you need to tick it to opt out.

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Privacy Suckering

Privacy Zuckering

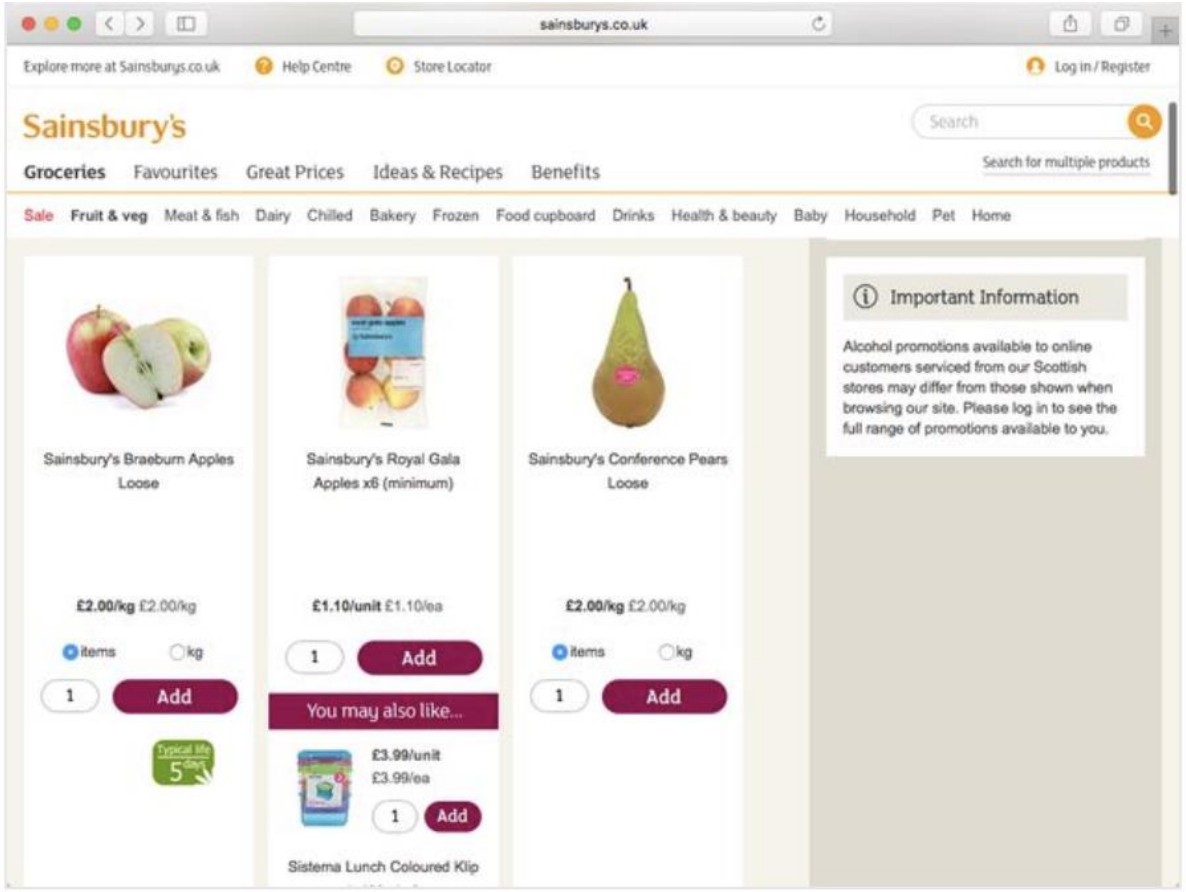
- You are tricked into publicly sharing more information about yourself than you really intended to. Named by **Tim Jones** as a homage to Facebook CEO Mark **Zucker**berg.
- In its early days, Facebook had a reputation for making it difficult for users to control their privacy settings, and generally making it very easy to "overshare" by mistake. In response to feedback from consumers and privacy groups, Facebook has created a clearer, easier to use privacy settings area.



Price Comparison Prevention

Price Comparison Prevention

- The retailer makes it hard for you to compare the price of an item with another item, so you cannot make an informed decision.



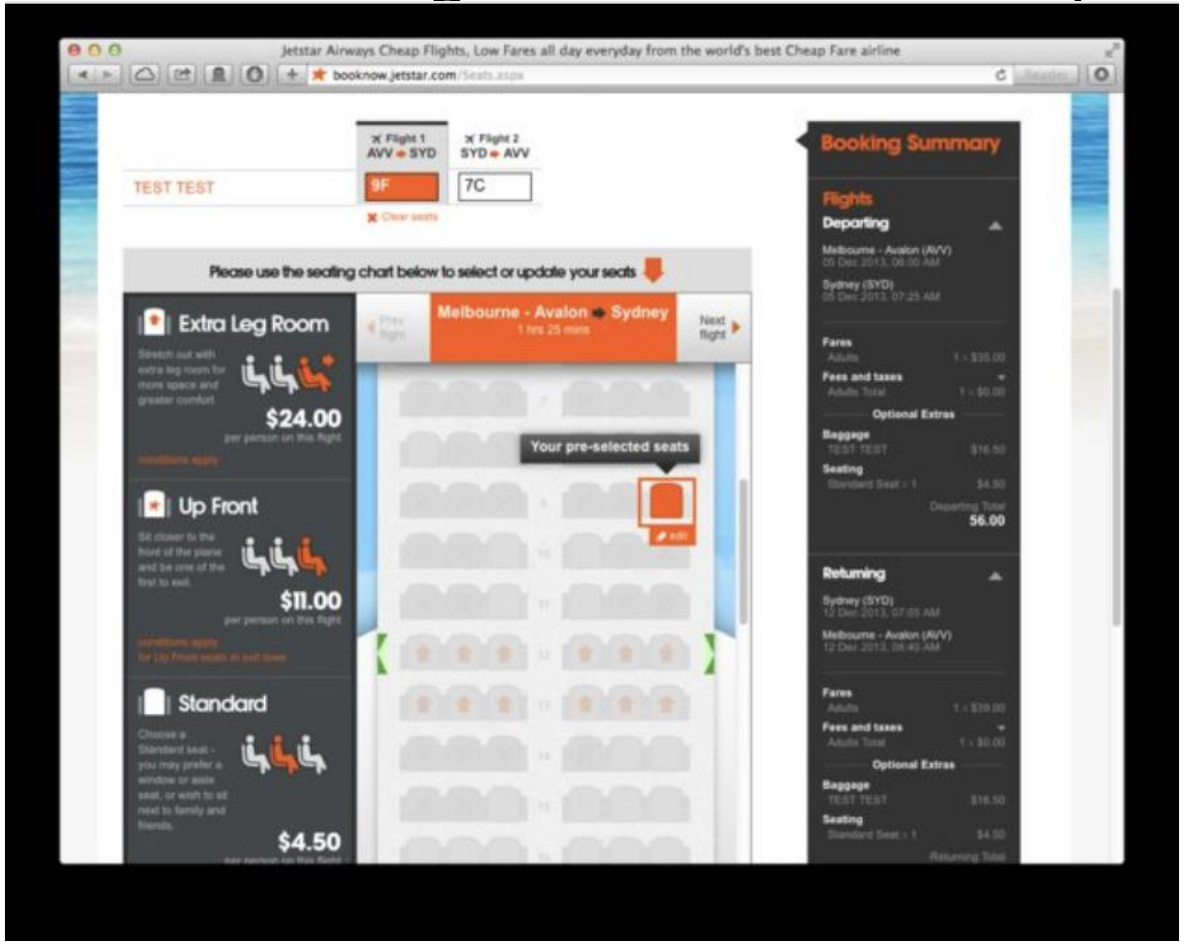
- Retailers typically achieve this by creating different bundles where it is not easy work out the unit price of the items within the bundles.
- In the Sainsbury's website, it is not possible to compare the price of the first item to the second item.



Misdirection

Misdirection

- The design purposefully focuses your attention on one thing in order to distract you attention from another.



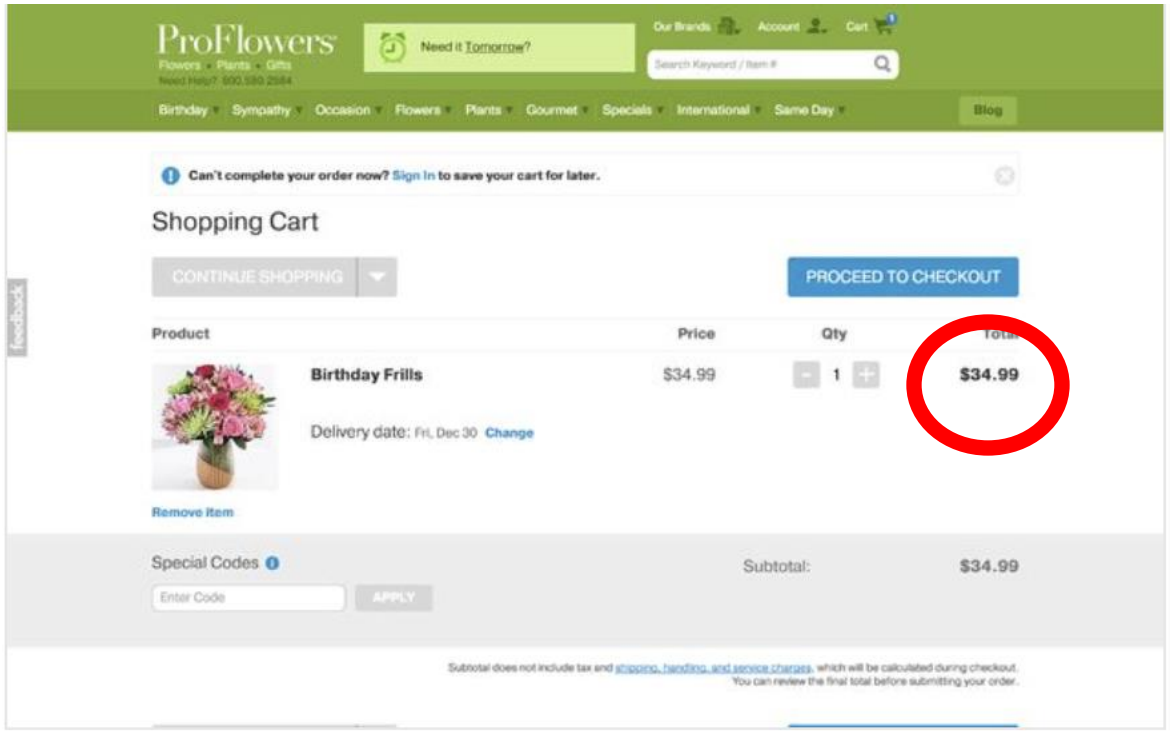
- On **jetstar.com**, when this page loads, they have already preselected a random seat for you.
- They use word-play trickery, as a "preselected" random seat here costs $5, but if you opt out via the tiny "skip seat selection" link at the bottom of the page, you're assigned a random seat without the $5 charge.

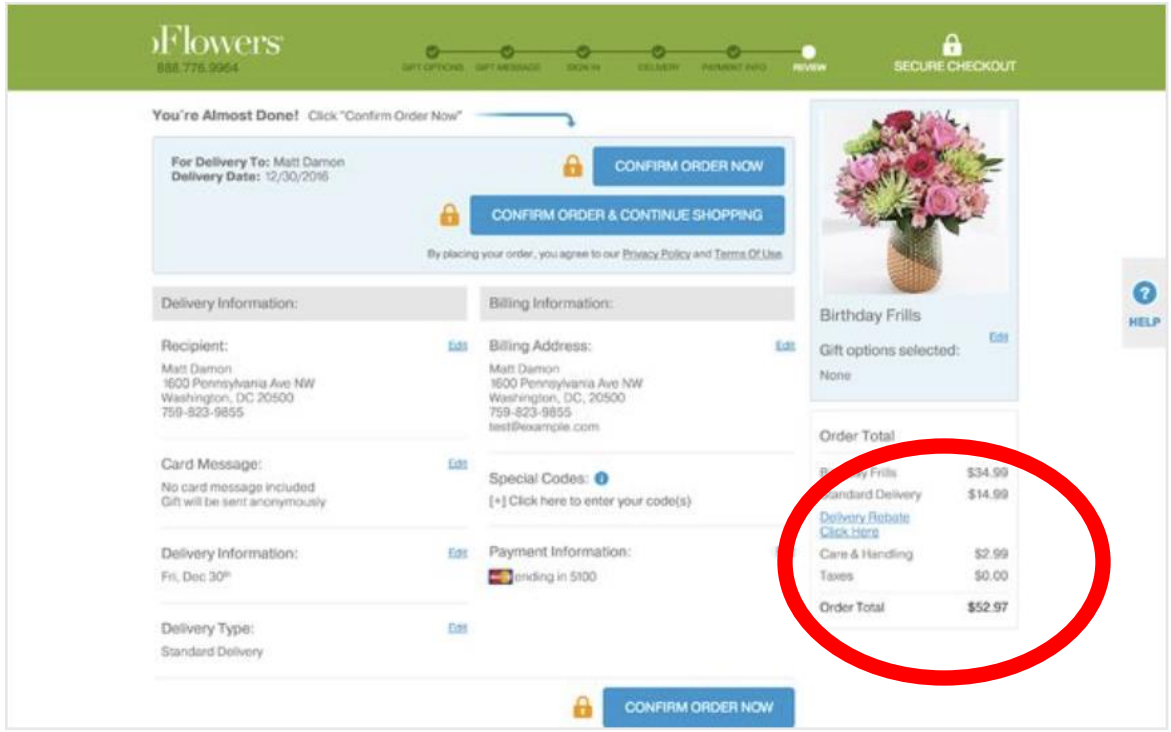


Hidden Costs

Hidden Costs

- You get to the last step of the checkout process, only to discover some unexpected charges have appeared, e.g. delivery charges, tax, etc.







Bait & Switch

Bait & Switch

- The user sets out to do one thing, but a different, undesirable thing happens instead.
- The most famous example of digital bait and switch was Microsoft's misguided approach to getting people to upgrade their computers to Windows 10.



Bait & Switch

- They switched the meaning of the “X” button at the top right to mean the opposite of what it normally means. In all other versions of Windows going back to the 1980s, this button means “close”. In this specific instance, they changed it to mean “Yes, I do want to upgrade my computer to Windows 10”.





Confirmshaming

Confirmshaming

- Confirmshaming is the act of guilting the user into opting in to something. The option to decline is worded in such a way as to shame the user into compliance.



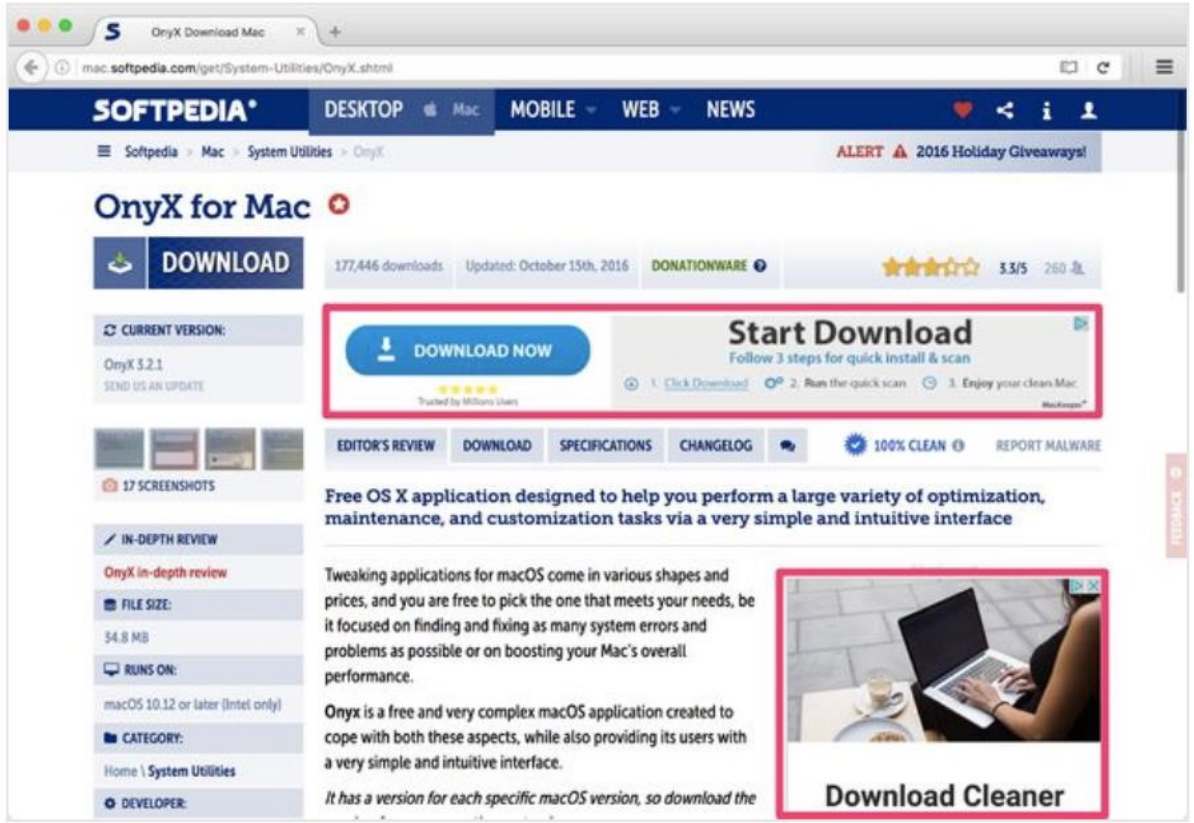
- Amazon uses confirmshaming to guilt users and discourage them from opting out.



Disguised Ads

Disguised Ads

- Adverts that are disguised as other kinds of content or navigation, in order to get you to click on them.



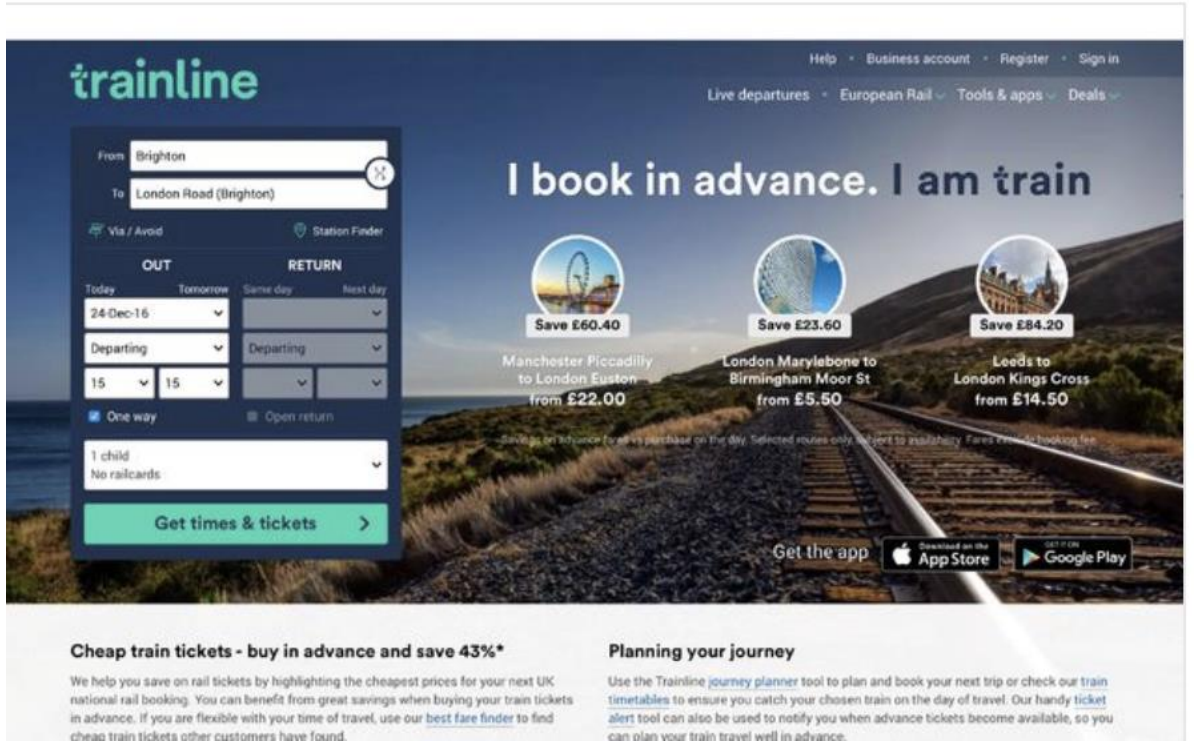
- Softpedia often run advertisements that look like a download button, tricking users into clicking on the ads rather than getting the thing they wanted.
- The real download link is at the top left of the page. The disguised ads are highlighted in red.



Forced Continuity

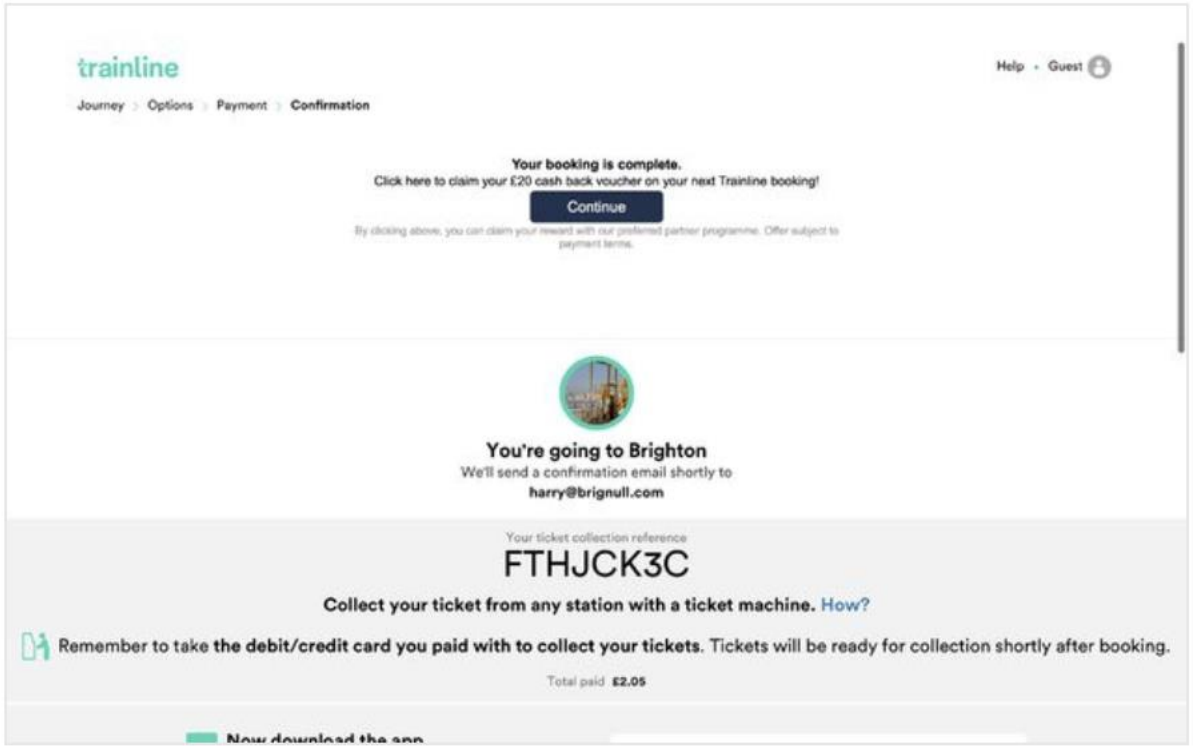
Forced Continuity

- When your free trial with a service comes to an end and your credit card silently starts getting charged without any warning. You are then not given an easy way to cancel the automatic renewal.
- thetrainline.com allows you to buy a ticket in the normal way.



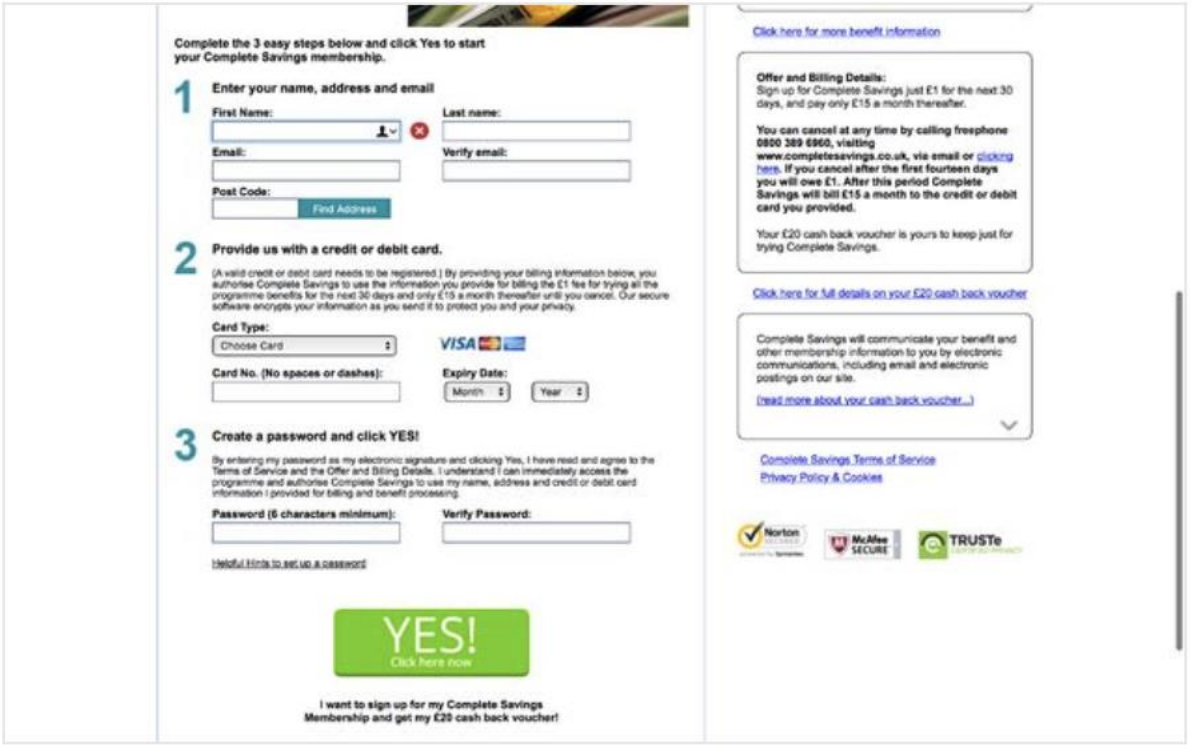
Forced Continuity

- When you complete the process, you are taken to a confirmation page, as shown below. The top box of content on the page is a **disguised ad**, using the same font and the same dark blue button style used by thetrainline.com.
- Having clicked a continue button a few times during the checkout process, you're very likely to think you need to click this button too, just to finish your booking. In fact, this isn't true.



Forced Continuity

- When you click the continue button, you're taken to another site (one-time-offer.com), which tries to convince you to sign up for a £20 money-off voucher.
- If you sign up to this service, you are enrolled into a £15 a month subscription which continues forever



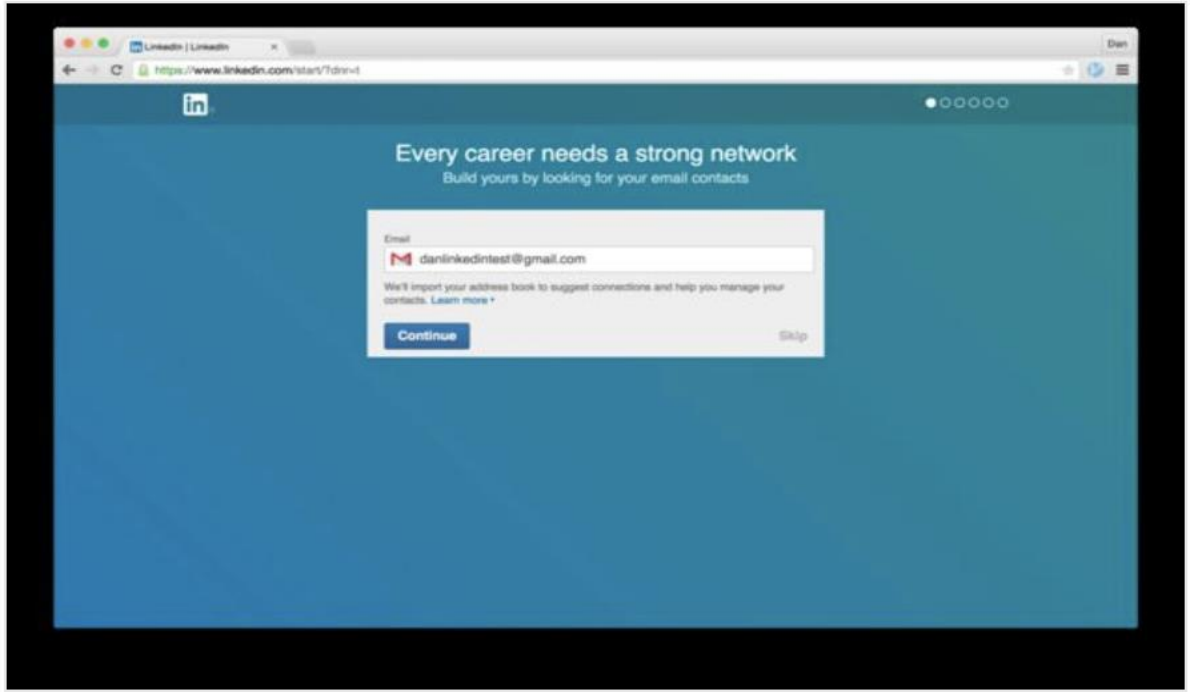
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DARK PATTERN

Friend Spam

Friend Spam

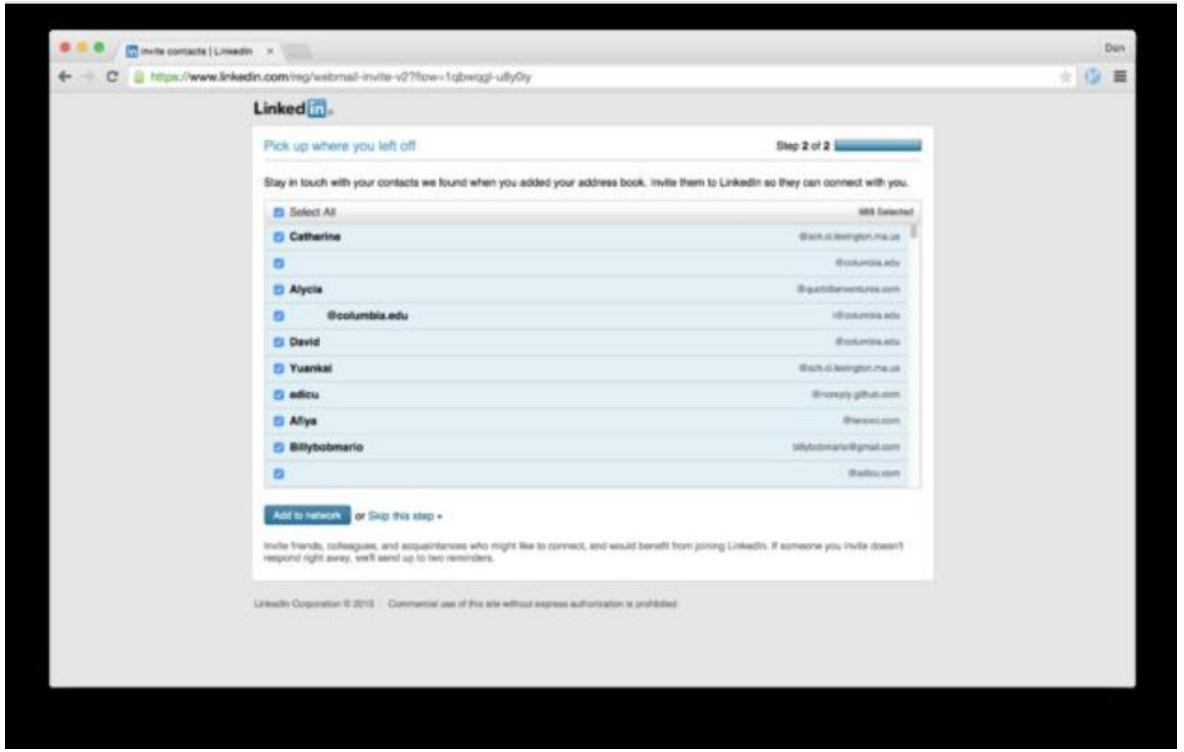
- The product asks for your email or social media permissions under the claim it will be used for a desirable outcome (e.g. finding friends), but then spams all your contacts in a message that claims to be from you.



- As part of the sign-up process to Linkedin, they encourage you to give them access to your email account, on the premise that it will give "your career a strong network" (see below).

Friend Spam

If you allow yourself to be guided through the steps, you're taken to the page shown below. If you click the innocent-sounding button "Add to network" you are inadvertently giving Linkedin permission to spam every person you've ever emailed.



Dark Patterns:
Research Topics

Research Topics

- Meta-analysis
- GDPR
- Intelligent Systems
- Mobile Apps
- Cookie Consent

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DARK PATTERNS

Meta-Analysis

DARK PATTERNS: Mathur *et al*. (2019)

- Mathur, A., Acar, G., Friedman, M.J., Lucherini, E., Mayer, J., Chetty, M. and Narayanan, A., 2019. *Dark patterns at scale: Findings from a crawl of 11K shopping websites*. Proceedings of the ACM on Human-Computer Interaction, 3(CSCW), pp.1-32.

DARK PATTERNS: Mathur *et al*. (2019)

- Mathur *et al*. (2019) did a meta-analysis of 11,286 shopping websites, and created a taxonomy to try to explain how dark patterns affects user decision-making by exploiting cognitive biases.

DARK PATTERNS: Mathur *et al*. (2019)

- They identified the following cognitive biases being used:
  - *The Anchoring Effect*
  - *The Bandwagon Effect*
  - *The Default Effect*
  - *The Framing Effect*
  - *The Scarcity Bias*
  - *The Sunk Cost Fallacy*

DARK PATTERNS: Mathur *et al*. (2019)

- Their taxonomy have the following characteristics:
  - *Asymmetric*
  - *Covert*
  - *Deceptive*
  - *Hides Information*
  - *Restrictive*

DARK PATTERNS: Mathur *et al*. (2019)

- They found that 11.1% (1254 websites) of the sites had dark patterns, and recommend the development of plug-ins for browsers to help detect these patterns.

DARK PATTERNS: Mathur *et al*. (2019)

- They also found that many of these patterns are unlawful in the United States law (under Section 5 of the Federal Trade Commission Act and similar state laws), and in the European Union, under the Unfair Commercial Practices Directive.



G.D.P.R.

DARK PATTERNS: Nouwens *et al*. (2020)

- Nouwens, M., Liccardi, I., Veale, M., Karger, D. and Kagal, L., 2020, April. *Dark patterns after the GDPR: Scraping consent pop-ups and demonstrating their influence*. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (pp. 1-13).

DARK PATTERNS: Nouwens *et al*. (2020)

- Nouwens *et al*. discuss the growth of Consent Management Platforms (CMPs) which are software systems that manage the interact between user and websites of organization's web sites, recording (and updating) their privacy preferences, and getting consent for recording interactions with cookies.

DARK PATTERNS: Nouwens *et al*. (2020)

- Crucially these CMPs are compliant with GDPR (the General Data Protection Regulation) however it is still possible for a website to employ Dark Patterns to circumvent GDPR, and almost 90% of the sites with CMPs surveyed were in some way breaching GDPR.



DARK PATTERNS

Intelligent Systems

DARK PATTERNS: Chromik *et al*. (2019)

- Chromik, M., Eiband, M., Völkel, S.T. and Buschek, D., 2019, March. *Dark Patterns of Explainability, Transparency, and User Control for Intelligent Systems*. In IUI Workshops (Vol. 2327).

DARK PATTERNS: Chromik *et al*. (2019)

- Chromik, *et al*. (2019) explore how there is potential for dark patterns to be used in Intelligent Systems.
- An intelligent system is computer system with an embedded artificial intelligence that can work to solve well-defined tasks, e.g. object recognition, medical diagnosis, language translation.

DARK PATTERNS: Chromik *et al*. (2019)

- As a consequence of GDPR, these systems must be able to provide some explanation as to how they came to specific decisions.
- Some intelligent systems incorporate *explanation facilities* to support users in understanding decisions.

DARK PATTERNS: Chromik *et al*. (2019)

- However this paper discusses the possibility of Intelligent Systems using Dark Patterns in conveying these explanations to get further data from the users.

DARK PATTERNS: Chromik *et al*. (2019)

- For example, the system could use the *Forced Data* Dark Pattern to collect valuable user data under the pretext of explanation. So the user might be forced to provide additional personal information (e.g., social connections) before receiving personalized explanations. Otherwise, the user would be left off with a generic high-level explanation

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DARK PATTERNS

Mobile Apps

DARK PATTERNS: Di Geronimo *et al*. (2020)

- Di Geronimo, L., Braz, L., Fregnan, E., Palomba, F. and Bacchelli, A., 2020, *UI Dark Patterns and Where to Find Them: A Study on Mobile Applications and User Perception*.

DARK PATTERNS: Di Geronimo *et al*. (2020)

- Di Geronimo, *et al*. (2020) explore the use of Dark Patterns in mobile apps.
- They looked at 240 popular mobile apps and explored whether or not these apps included any dark patterns.

DARK PATTERNS: Di Geronimo *et al*. (2020)

- Their analysis showed that 95% of the apps they reviewed included one or more Dark Patterns, with an average of 7.4 malicious designs per app, with a standard deviation of 5.

DARK PATTERNS: Di Geronimo *et al*. (2020)

- Almost 10% of the apps included 0, 1, or 2 Dark Patterns (N=33), 37% of the apps contained between 3 to 6 Dark Patterns (N=89), while the remaining 49% included 7 or more (N=118).

DARK PATTERNS: Di Geronimo *et al*. (2020)

- They also conducted an online experiment with 589 users on how they perceive Dark Patterns in such apps.

DARK PATTERNS: Di Geronimo *et al*. (2020)

- Overall, the majority of our users did not spot malicious designs in the app containing Dark Patterns (55%), some were unsure (20%), and the remaining found a malicious design in the app (25%)

DARK PATTERNS: Di Geronimo *et al*. (2020)

- But they found that most users did perform better in recognizing malicious designs if informed on the issue.

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DARK PATTERNS

Cookie Consent

DARK PATTERNS: Grassl *et al*. (2020)

- Grassl, Paul & Schraffenberger, Hanna & Borgesius, Frederik & Buijzen, Moniek. (2020). *Dark and bright patterns in cookie consent requests*. 10.31234/osf.io/gqs5h.

DARK PATTERNS: Grassl *et al*. (2020)

- Grassl, *et al*. (2020) looked at cookie consent requests in the context of Dark Patterns to explore whether or not they undermine principles of EU privacy law.

DARK PATTERNS: Grassl *et al*. (2020)

- They undertook two online experiments where they investigated the effects of common design nudges on users' consent decisions and their perception of control over their personal data in these situations.

DARK PATTERNS: Grassl *et al*. (2020)

- In the first experiment (n = 228) they explored the effects of dark patterns to encourage the participants to select the privacy-unfriendly option, and the experiment revealed that most people agreed to all consent requests regardless of dark patterns.
- The research indicated that the dark patterns made no difference to the participants' behavior.

DARK PATTERNS: Grassl *et al*. (2020)

- The first experiment, also showed that despite generally low levels of perceived control, obstructing the privacy-friendly option led to more rather than less perceived control for the participants.

DARK PATTERNS: Grassl *et al*. (2020)

- In the second experiment (n = 255) the participants we presented with patterns to select the privacy-friendly option (bright patterns).
- The bright pattern did succeed in swaying people effectively towards the privacy-friendly option,

DARK PATTERNS: Grassl *et al*. (2020)

- The second experiment also looked at the perceived control of the participants, and it found that it stayed the same compared to Experiment 1.

DARK PATTERNS: Grassl *et al*. (2020)

- Overall the researchers concluded about Experiment 1 that whether the participants were presented with a dark pattern or not, they have been conditioned by years of practice to consent, and therefore they concluded that the EU's consent requirement for tracking cookies does not work as intended.

Discussion

Reviewing the patterns we have discussed, select the top 3 you think are most unethical

1. **Trick questions**
2. **Sneak into Basket**
3. **Roach Motel**
4. **Privacy Zuckering**
5. **Price Comparison Prevention**
6. **Misdirection**
7. **Hidden Costs**
8. **Bait and Switch**
9. **Confirmshaming**
10. **Disguised Ads**
11. **Forced Continuity**
12. **Friend Spam**

DARK PATTERNS: Activity

- Discuss one the following themes:

	Discussion Card ***Some Dark Patterns are worse than others.***	**Discussion Card** ***It's the users' own fault if they fall victim to a dark pattern.***
Discussion Card ***I rarely read everything when interacting with a site and have been caught out by the interface.***	**Discussion Card** ***If I was asked to write an interface with a dark pattern, I wouldn't do it, even for €1 million.***	**Discussion Card** ***Some organisations have plausible explanations why they have interfaces that use dark patterns.***